Summary in Graph

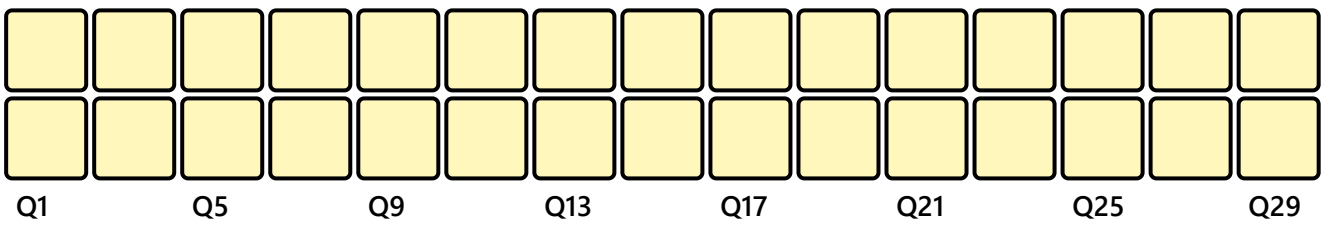
Exam Summary (GATE Overflow CS Test Series 2026 | Discrete Mathematics | Test 3)

Qs. Attempted:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)
Correct Marks:	0 (0 + 0)
Penalty Marks:	0 (0 + 0)
Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (3) out of 33 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Numerical Type | Award: 1 | Penalty: 0 | Set Theory & Algebra

Let $'a'$ be an element in a group such that $ord(a) = 2020$. The order of a^{43} is _____

Your Answer: | Correct Answer: 2020 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 Numerical Type | Award: 1 | Penalty: 0 | Set Theory & Algebra

Let G be a group with order 45, and H a non-abelian subgroup of G. Assuming $H \neq G$, what is the smallest possible order of H?

Your Answer: | Correct Answer: 15 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

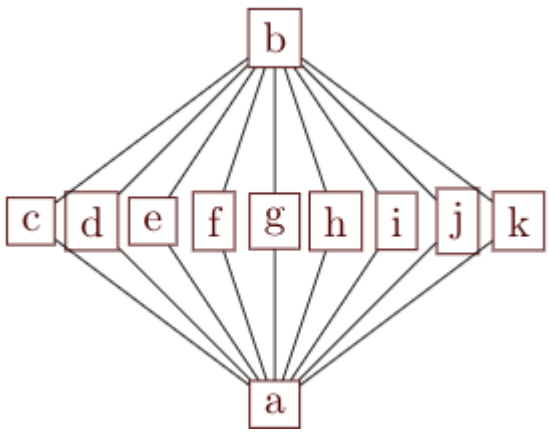
Let $G = \langle a \rangle$. The smallest subgroup of G containing a^{2020} and a^{1719} is _____

A. a^4
B. a^{1719}
C. a^{2020}
D. a^1

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Numerical Type | Award: 1 | Penalty: 0 | Set Theory & Algebra

Consider the below lattice.



The number of complements of element $'g'$ is _____

Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Multiple Select Type | Award: 1 | Penalty: 0 | Set Theory & Algebra

Which of the following is NOT correct? (Mark all the appropriate choices)

A. $(\mathbb{R}^*, \times)$, the set of real numbers excluding 0, under multiplication operation is a group.
B. $(\mathbb{Q}, \times)$, the set of rational numbers under multiplication operation is a group.
C. $(\mathbb{Z}, +)$, the set of integers under addition operation is a group.
D. $\mathbb{Z}_n$, the set of integers $\{0, 1, \ldots, n-1\}$, under addition modulo n operation is a group.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

What is the solution to the recurrence relation $a_n = a_{n-1} + n$ with $a_0 = 0$?

A. $a_n = \frac{n(n-1)}{2}$
B. $a_n = \frac{(n-1)(n+1)}{2}$
C. $a_n = \frac{(n-1)(n+1)}{4}$
D. $a_n = \frac{n(n+1)}{2}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

Consider a prime number p and a positive integer a such that p divides a. The remainder when a^{p-1} is divided by p is

A. 1
B. 0
C. $p-1$
D. Cannot be determined

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

Which of the following is/are correct? ("/" is the "divides" relation)

- $S1 : [\{1, 2, 3, 6, 9, 18, 36, 72\},\ /]$ is a lattice.
- $S2 : [\{1, 3, 9, 27, 81, 243\},\ /]$ is a total order.

A. Only $S1$
B. Only $S2$
C. Both $S1$ and $S2$
D. None of the above

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

Which of the following is/are correct?

I. A relation that is reflexive, anti-symmetric, and transitive is called a partial order.
II. When every two elements in a set are comparable, the relation is called a total ordering (linear ordering).
III. The poset $(\mathbb{Z}, \leq)$ is not totally ordered, where $\mathbb{Z}$ is the set of integers.
IV. The inclusion relation $\subseteq$ is a partial ordering on the power set of a set.

A. I and II only
B. I and III only
C. I, II and III only
D. I, II and IV only

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

Let $(G, *)$ be a group of order P, where P is a prime number. The number of subgroups of G is _____

A. 0
B. P^2
C. $P^2 - 2$
D. 2

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

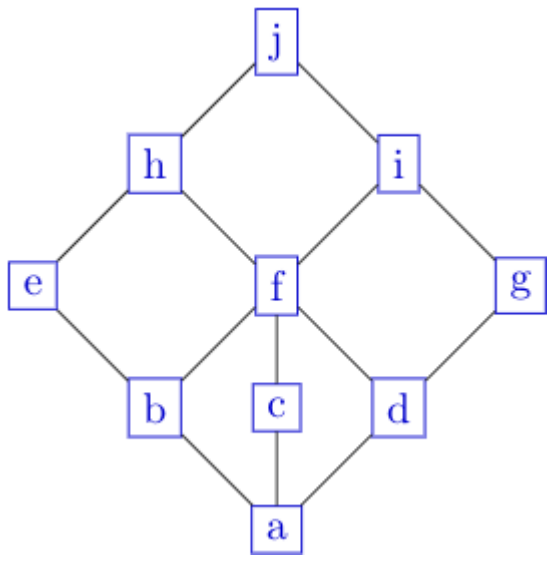
Q #11 Numerical Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

The value of x_5 for the recurrence relation defined by $x_n = 5x_{n-1} + 3$ with initial condition $x_1 = 3$ is _____

Your Answer: | Correct Answer: 2343 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

The following is the Hasse diagram of the poset $[\{a, b, c, d, e, f, g, h, i, j\}, \preceq]$



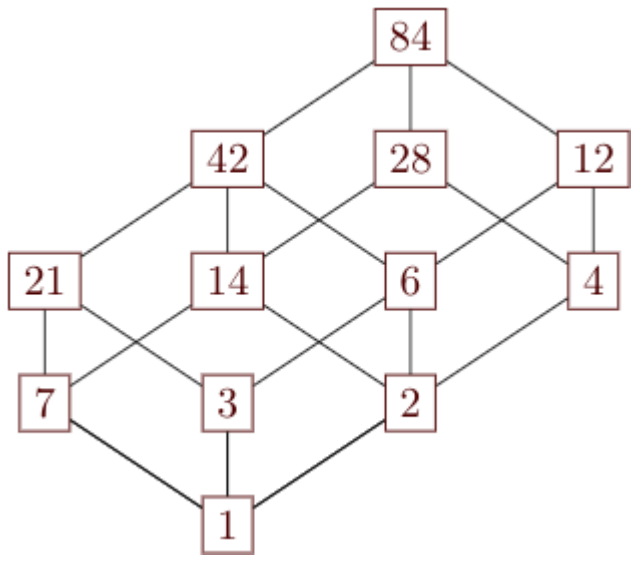
The poset is :

A. not a lattice.
B. a complemented and distributive lattice.
C. neither a complemented lattice nor a distributive lattice.
D. a distributive lattice but not a Boolean algebra.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Numerical Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

The following is the Hasse diagram of the poset $[\{1, 2, 3, 4, 6, 7, 12, 14, 21, 28, 42, 84\}, \mid]$, where $\mid$ is the "divides" relation.

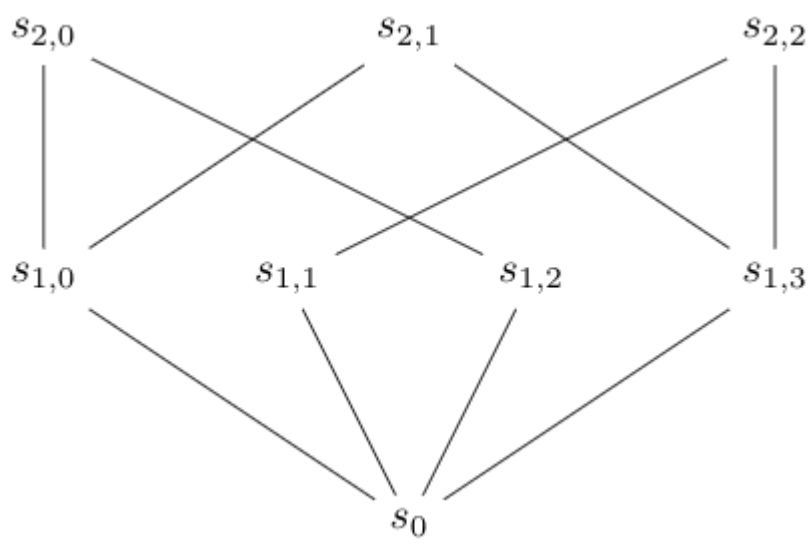


If the number of complements of $1, 3$ and 4 are α, β and γ respectively, then the value of $2\alpha + 3\beta + 4\gamma$ is _____

Your Answer: | Correct Answer: 9 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

The following Hasse diagram describes a



A. Join semi lattice
B. Meet semi lattice
C. Lattice
D. None of the above

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Which of the following is not correct regarding a bounded lattice?

A. $[\{1, 2, 3, 6, 9, 18\}, \mid]$ is a bounded lattice.
B. $[I, \leq]$ is not a bounded lattice, where I is the set of integers.
C. $[[0, 1], \leq]$ is bounded lattice.
D. $[(0, 1), \leq]$ is bounded lattice.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

The solution to the recurrence relation $a_n = a_{n-1} + 2^n$ with $a_0 = 1$ is _____

A. $a_n = 2^n - 1$
B. $a_n = 2^{n+1} - 2$
C. $a_n = 2^{n+1} - 1$
D. $a_n = 2^{n-1} - 1$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Which of the following is NOT correct? (Mark all the appropriate choices)

A. Any non single element group G has at least 2 subgroups.
B. Every subgroup of a cyclic group is cyclic.
C. A cyclic group is not abelian.
D. Any subgroup of an abelian group is also abelian.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

The explicit formula for the sequence defined by the recurrence relation $x_n = 2x_{n-1} + 15x_{n-2}$ subject to the initial conditions, $x_1 = 2, x_2 = 4$ is.

A. $\frac{(-3)^n+5^n}{4}$
B. $\frac{(-1)^{n+1}3^n+5^n}{4}$
C. $\frac{(-1)^n3^n-5^n}{4}$
D. $\frac{3^n-5^n}{4}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 Numerical Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Let G be a finite group on 243 elements. The size of the largest possible proper subgroup of G is _____

Your Answer: | Correct Answer: 81 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 Numerical Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

The remainder when 99^{96} is divided by 97 is _____

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 02sec | Discuss

Q #21 Numerical Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

How many non-isomorphic abelian groups of order 32 are there?

Your Answer: | Correct Answer: 7 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 Numerical Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Let G be a cyclic group of order 96, then the number of subgroups of G is _____

Your Answer: | Correct Answer: 12 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 Numerical Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

If G is cyclic group of order 48, then how many element of order 8 are in G?

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

How many ways can we fill the following table so that the binary operation $*$ is commutative?

$*$	**a**	**b**	**c**	**d**
a	d	c		a
b		d	a	
c	b		d	c
d		b		d

A. 2
B. 3
C. 1
D. 4

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 Numerical Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

The remainder when 60^{61} is divided by 61 is _____

Your Answer: | Correct Answer: 60 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

If $G = \{1, 2, 3, 4, 5, 6\}$ is a group with respect to $\otimes_7$ (multiplication modulo 'm'), then which of the following is NOT true? (Mark all the appropriate choices)

A. The inverse of 1 is 1
B. The inverse of 6 is 3
C. The inverse of 2 is 2
D. The inverse of 3 is 5

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Suppose G is a group that has exactly 48 distinct elements of order 5. How many distinct subgroups of order 5 does G have?

A. 12
B. 6
C. 1
D. 4

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

What is the solution to the recurrence relation

$$f_n = f_{n-1} + f_{n-2} \quad \text{with } f_0 = 0, f_1 = 1?$$

A. $f_n = \frac{1}{\sqrt{5}}\left(\frac{(1+\sqrt{5})}{2}\right)^n - \frac{-1}{\sqrt{5}}\left(\frac{(1-\sqrt{5})}{2}\right)^n$
B. $f_n = \frac{1}{\sqrt{5}}\left(\frac{(1+\sqrt{5})}{2}\right)^n + \frac{-1}{\sqrt{5}}\left(\frac{(1-\sqrt{5})}{2}\right)^n$
C. $f_n = \frac{1}{\sqrt{5}}\left(\frac{(1-\sqrt{5})}{2}\right)^n + \frac{-1}{\sqrt{5}}\left(\frac{(1+\sqrt{5})}{2}\right)^n$
D. $f_n = \frac{1}{\sqrt{5}}\left(\frac{(1-\sqrt{5})}{2}\right)^n - \frac{-1}{\sqrt{5}}\left(\frac{(1+\sqrt{5})}{2}\right)^n$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 Numerical Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Calculate $a_{100} + a_{99}$ where a_n is given by the recurrence relation $a_n = 2a_{n-1} - a_{n-2}$ with $a_0 = 5, a_1 = 4$?

Your Answer: | Correct Answer: -189 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

The number of maximal and minimal elements for the poset $[\ \{2, 4, 6, 9, 12, 18, 27, 36, 48, 60, 72\},\ |\ ]$, where $|$ is the "divides" relation is

A. $2, 1$
B. $4, 1$
C. $3, 2$
D. $4, 2$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

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